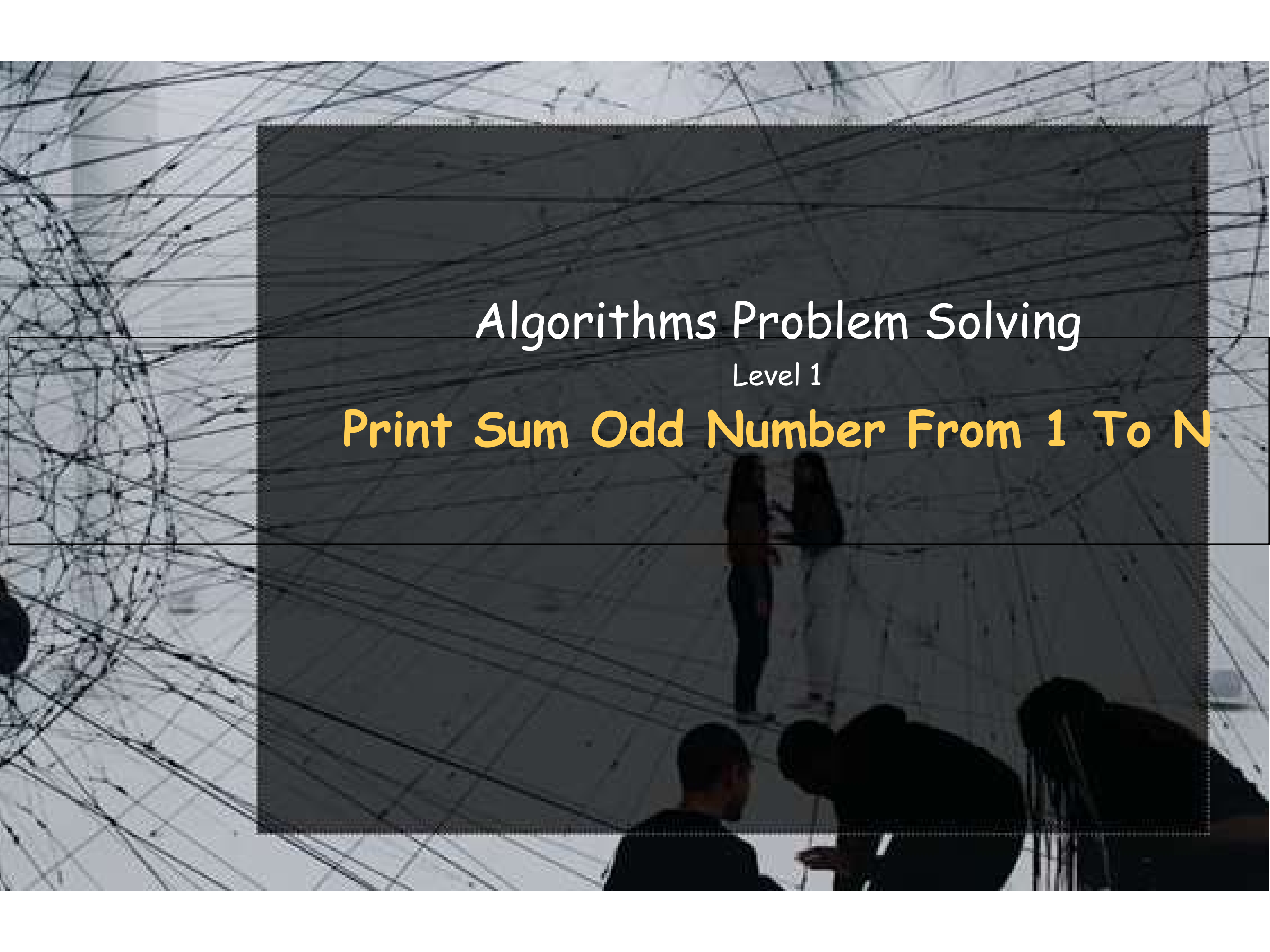


Algorithms Problem Solving

Level 1

Print Sum Odd Number
From 1 To N



Algorithms Problem Solving

Level 1

Print Sum Odd Number From 1 To N

🔍 Definition

- ◆ This program reads a number N and calculates the sum of all odd numbers from 1 to N .
- ◆ It uses an `enum` to represent whether a number is odd or even.
- ◆ The program checks each number in the range and adds only the odd numbers to the sum.

Explanation

- ◆ ``enOddOrEven`` is an enumeration with two values:
 - ``Odd = 1``
 - ``Even = 2``
- ◆ ``readNumb()`` reads N from the user and returns it.
- ◆ ``checkOddOrEven()`` uses the ``%`` operator to determine whether a number is odd or even.
- ◆ ``num % 2 != 0`` means the number has a remainder when divided by 2, so it is odd.
- ◆ ``sumOddNumbersFrom1ToN()`` starts ``sum`` at 0 and loops from 1 to N.
- ◆ For each number, it calls ``checkOddOrEven()``.
- ◆ If the result is ``Odd``, the number is added to ``sum``.
- ◆ Finally, the function returns the total sum.
- ◆ ``sumOddNumbersFrom1ToN(readNumb())`` reads N and passes it directly to the summation function.

Steps in main()

1. Read the number N.
2. Set `sum` to 0.
3. Start the counter at 1.
4. Check whether the current number is odd.
5. If it is odd, add it to `sum`.
6. Increase the counter by 1.
7. Repeat until the counter reaches N.
8. Return the final sum.
9. Print the result.

💡 Example

◆ If the user enters:

$N = 10$

◆ The odd numbers from 1 to 10 are:

1, 3, 5, 7, 9

◆ The program calculates:

$1 + 3 + 5 + 7 + 9 = 25$

➡ Result

Sum odd numbers = 25

Note

◆ The `%` operator returns the remainder of an integer division.

◆ For example:

$7 \% 2 = 1 \rightarrow \text{Odd}$

$8 \% 2 = 0 \rightarrow \text{Even}$

⊘ Warning

- ◆ If N is less than 1, the loop does not execute and the result remains 0.



Comparison

Odd Number

- ✓ ``num % 2 != 0``
- ✓ Has a remainder when divided by 2.
- ✓ Examples: 1, 3, 5, 7, 9.

Even Number

- ✓ ``num % 2 == 0``
- ✓ Has no remainder when divided by 2.
- ✓ Examples: 2, 4, 6, 8, 10.

✦✦ Summary

- ✓ The program calculates the sum of odd numbers from 1 to N.
- ✓ ``% 2`` is used to determine whether a number is odd or even.
- ✓ The ``enum`` ``enOddOrEven`` represents the two possible results.
- ✓ Only numbers classified as ``Odd`` are added to ``sum``.
- ✓ The program uses separate functions for reading, checking, and calculating.

Thank you 😊